

Things You Wanted to Know!

Q1. What are the advantages of using microcontroller (MCU) development boards like Arduino, Raspberry Pi and Beagle Bone?

Pamarthi Kanakaraja

Asst. professor, Andhra Pradesh
Through email

A1. An MCU is a standalone single-chip integrated circuit that contains a central processing unit, read only memory (ROM) to store the program, random access memory (RAM) to store variables used in the execution of the program and various input/output (I/O) buses like SPI, I2C and UART to connect to the outside world.

As compared to MCUs, an Arduino simplifies the amount of hardware and software development needed to get a system functioning. Arduino platform is designed to give the user complete control of hardware. On software side, it provides a number of libraries to make programming the Arduino easier. Simplest of these are functions to control and read I/O pins rather than having to fiddle with bus/bit masks normally used to interface with Atmega I/O. By using Arduino integrated development environment, you can write programs that can interface with almost limit-less hardware including switches, sensors, LCDs, other MCUs, the Internet and so on. Arduino hardware and software is open source, and the designs and software can be used to make projects.

Devices like Raspberry Pi and Beagle Board are designed to function on a much higher level. With already integrated hardware that takes care of things like the ethernet, video and audio processing, large quantities of RAM and an almost unlimited amount of storage space, these are really mini-computers. You can run complete operating systems (OSes) like Linux and Android and develop programs

within those OSes that can control system functions and I/Os that are made available.

Q2. What is the difference between Bluetooth module (BT 24, BT 53 and HC 05) and Bluetooth dongle?

Sachin Ambesange

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Through email

A2. Bluetooth is a standardised protocol for sending and receiving data via a 2.4GHz wireless link. It is a secure protocol and is perfect for short-range, low-power, low-cost, wireless transmissions between electronic devices.

Bluetooth modules (Fig. 1) like BT 24 and BT 53 are Tx-Rx pair modules from Amp'ed radio frequency technology. These are used in applications that require basic Bluetooth functionality. Their hardware is based



Fig. 1: Bluetooth module



Fig. 2: Bluetooth dongle

on Cortex-M3 and Cortex-M4 micro-processors, respectively. Their radio classification is Class 2 and Class 1, respectively. Radio classification is based on the range of transmission and maximum transmitted power. Maximum range of transmission in Class 1 is about 100m, whereas the range is limited to 10m in case of Class 2 radio. Both are SPI and I2C compatible.

HC-05 module (Fig. 2) is Bluetooth serial port protocol module, designed for transparent wireless serial connection setup. Serial port Bluetooth module is a fully-qualified Bluetooth V2.0 + EDR (enhanced data rate) having 3Mbps modulation with complete 2.4GHz radio transceiver and baseband. It uses CSR Bluecore 04-External single-chip Bluetooth system with complementary metal-oxide semiconductor technology and with

adaptive frequency hopping feature. It has a footprint of 12.7mm x 27mm. It aims to simplify the overall design/development cycle.

So bluetooth modules can be used as electronic devices by the user as per requirement.

Bluetooth dongles are USB network adaptors that allow a computer to communicate with Bluetooth devices such as mobile phones, mice, keyboards, remotes and headsets. Dongles are meant for short-range wireless networking, typically within 10m (33-feet) or less. If you move too far from the dongle, connection to the computer either becomes slow or unavailable.

Q3. Is it possible to make an induction heater just to produce enough heat to cook food by modifying the circuit of a household inverter? If possible, please send the procedure and modifications to make it.

Soumik Dutta

Dibrugarh, Assam

Through email

A3. Yes, it is possible! A power inverter is required to change DC power from the battery, which is usually 12V or 24V, to conventional mains alternating current AC power at 230V. This means that you can use the power inverter to operate all kinds of domestic electrical appliances including an induction heater.

Normally, an induction heater/cook-top consumes energy in the range of 1.8kW to 2kW. Most household inverters are capable of producing this required energy. Simply connect the inverter to a battery and plug the induction heater into the inverter. The inverter draws its power from the battery, which needs to be recharged. The battery can be recharged by a generator, solar panels or a battery charger plugged into an AC outlet.

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